

Table I. Mean predictive performance of the Logistic Regression model with one independent variable

n	Handling Technique	IR	MLE				Score			
			Acc	Sens	Spec	BA	Acc	Sens	Spec	BA
200	Original	90:10	0.9192	0.1939	0.9317	0.5969	0.9279	0.4561	0.9802	0.7182
		80:20	0.8837	0.5436	0.9687	0.7561	0.8964	0.6733	0.9522	0.8128
		70:30	0.8674	0.7621	0.9126	0.8373	0.8675	0.7614	0.9130	0.8372
	RUS	90:10	0.8240	0.9018	0.8153	0.8586	0.8425	0.8852	0.8378	0.8615
		80:20	0.8460	0.9081	0.8305	0.8693	0.8622	0.8814	0.8574	0.8694
		70:30	0.8492	0.8917	0.8309	0.8613	0.8592	0.8717	0.8538	0.8628
	ROS	90:10	0.8291	0.9003	0.8212	0.8608	0.8528	0.8787	0.8499	0.8643
		80:20	0.8484	0.9103	0.8329	0.8716	0.8658	0.8837	0.8613	0.8725
		70:30	0.8500	0.8946	0.8308	0.8627	0.8577	0.8724	0.8513	0.8619
	SMOTE	90:10	0.8387	0.8960	0.8323	0.8641	0.8707	0.8555	0.8724	0.8640
		80:20	0.8495	0.9103	0.8343	0.8723	0.8698	0.8808	0.8671	0.8740
		70:30	0.8588	0.8834	0.8482	0.8658	0.8630	0.8579	0.8652	0.8616
	ANS	90:10	0.8371	0.8952	0.8306	0.8629	0.8651	0.8633	0.8653	0.8643
		80:20	0.8495	0.9090	0.8346	0.8718	0.8696	0.8809	0.8667	0.8738
		70:30	0.8585	0.8816	0.8486	0.8651	0.8636	0.8576	0.8662	0.8619
	DBSMOTE	90:10	0.8430	0.8863	0.8381	0.8622	0.8856	0.8182	0.8931	0.8556
		80:20	0.8538	0.8981	0.8428	0.8704	0.8810	0.8502	0.8886	0.8694
		70:30	0.8592	0.8754	0.8522	0.8638	0.8654	0.8383	0.8770	0.8576
500	Original	90:10	0.9068	0.0821	0.9497	0.5403	0.9245	0.4487	0.9774	0.7130
		80:20	0.8776	0.5161	0.9679	0.7420	0.8848	0.6386	0.9464	0.7925
		70:30	0.8643	0.7544	0.9113	0.8329	0.8640	0.7505	0.9127	0.8316
	RUS	90:10	0.8302	0.9051	0.8219	0.8635	0.8466	0.8845	0.8424	0.8634
		80:20	0.8379	0.8912	0.8246	0.8579	0.8504	0.8742	0.8445	0.8593
		70:30	0.8388	0.8865	0.8183	0.8524	0.8472	0.8602	0.8416	0.8509
	ROS	90:10	0.8329	0.9051	0.8249	0.8650	0.8426	0.8792	0.8385	0.8589
		80:20	0.8371	0.8886	0.8242	0.8564	0.8584	0.8707	0.8553	0.8630
		70:30	0.8398	0.8914	0.8177	0.8546	0.8566	0.8736	0.8494	0.8615
	SMOTE	90:10	0.8328	0.9057	0.8247	0.8652	0.8529	0.8743	0.8505	0.8624
		80:20	0.8376	0.8904	0.8244	0.8574	0.8509	0.8686	0.8464	0.8575
		70:30	0.8443	0.8733	0.8318	0.8526	0.8538	0.8455	0.8574	0.8515
	ANS	90:10	0.8329	0.9045	0.8249	0.8647	0.8543	0.8455	0.8581	0.8518
		80:20	0.8378	0.8908	0.8246	0.8577	0.8516	0.8662	0.8479	0.8570
		70:30	0.8443	0.8733	0.8318	0.8526	0.8543	0.8455	0.8581	0.8518
	DBSMOTE	90:10	0.8395	0.8965	0.8332	0.8648	0.8747	0.8311	0.8795	0.8553
		80:20	0.8435	0.8829	0.8337	0.8583	0.8619	0.8424	0.8667	0.8546
		70:30	0.8467	0.8676	0.8378	0.8527	0.8604	0.8281	0.8742	0.8512

Table I. (contd.)

n	Handling Technique	IR	MLE				SCORE			
			Acc	Sens	Spec	BA	Acc	Sens	Spec	BA
1,000	Original	90:10	0.9056	0.0616	0.9766	0.5305	0.9251	0.4503	0.9779	0.7141
		80:20	0.8787	0.5083	0.9713	0.7397	0.8878	0.6390	0.9501	0.7945
		70:30	0.8669	0.7526	0.9159	0.8342	0.8672	0.7583	0.9138	0.8360
	RUS	90:10	0.8340	0.8949	0.8272	0.8610	0.8502	0.8715	0.8478	0.8597
		80:20	0.8391	0.8932	0.8256	0.8594	0.8503	0.8720	0.8449	0.8584
		70:30	0.8473	0.8971	0.8259	0.8615	0.8537	0.8732	0.8453	0.8593
	ROS	90:10	0.8333	0.8952	0.8264	0.8608	0.8510	0.8718	0.8487	0.8602
		80:20	0.8388	0.8923	0.8254	0.8588	0.8517	0.8709	0.8468	0.8589
		70:30	0.8482	0.8951	0.8281	0.8616	0.8556	0.8733	0.8481	0.8607
	SMOTE	90:10	0.8335	0.8962	0.8265	0.8614	0.8523	0.8699	0.8503	0.8601
		80:20	0.8396	0.8926	0.8263	0.8595	0.8519	0.8693	0.8476	0.8584
		70:30	0.8523	0.8780	0.8413	0.8597	0.8587	0.8573	0.8594	0.8583
	ANS	90:10	0.8335	0.8959	0.8266	0.8612	0.8537	0.8671	0.8522	0.8596
		80:20	0.8395	0.8924	0.8262	0.8593	0.8527	0.8660	0.8494	0.8577
		70:30	0.8528	0.8789	0.8416	0.8603	0.8589	0.8558	0.8602	0.8580
	DBSMOTE	90:10	0.8404	0.8855	0.8358	0.8621	0.8720	0.8309	0.8766	0.8537
		80:20	0.8445	0.8831	0.8349	0.8590	0.8649	0.8357	0.8722	0.8540
		70:30	0.8533	0.8711	0.8457	0.8584	0.8599	0.8381	0.8693	0.8537

Table II. Mean predictive performance of the Logistic Regression model with three independent variables

n	Handling Technique	IR	MLE				Score			
			Acc	Sens	Spec	BA	Acc	Sens	Spec	BA
200	Original	90:10	0.9222	0.2232	0.9438	0.6115	0.9402	0.5842	0.9798	0.7820
		80:20	0.9027	0.6200	0.9733	0.7967	0.9150	0.7549	0.9550	0.8550
		70:30	0.8960	0.8352	0.9220	0.8786	0.8965	0.8224	0.9283	0.8754
	RUS	90:10	0.8402	0.9378	0.8293	0.8836	0.8679	0.9262	0.8614	0.8938
		80:20	0.8592	0.9465	0.8373	0.8919	0.8826	0.9224	0.8727	0.8976
		70:30	0.8710	0.9492	0.8375	0.8933	0.8865	0.9134	0.8750	0.8942
	ROS	90:10	0.8530	0.9520	0.8419	0.8970	0.8946	0.9217	0.8916	0.9066
		80:20	0.8677	0.9493	0.8473	0.8983	0.8946	0.9133	0.8899	0.9016
		70:30	0.8728	0.9510	0.8392	0.8951	0.8889	0.9128	0.8786	0.8957
	SMOTE	90:10	0.8402	0.9378	0.8293	0.8836	0.9050	0.8887	0.9068	0.8977
		80:20	0.8592	0.9465	0.8373	0.8919	0.9002	0.9070	0.8985	0.9028
		70:30	0.8710	0.9492	0.8375	0.8933	0.8937	0.8918	0.8945	0.8931
	ANS	90:10	0.8588	0.9387	0.8499	0.8943	0.9089	0.8713	0.9131	0.8922
		80:20	0.8690	0.9516	0.8484	0.9000	0.9004	0.9013	0.9002	0.9007
		70:30	0.8793	0.9429	0.8521	0.8975	0.8944	0.8903	0.8962	0.8932
	DBSMOTE	90:10	0.8612	0.9380	0.8527	0.8954	0.9123	0.8645	0.9176	0.8911
		80:20	0.8711	0.9479	0.8518	0.8999	0.9052	0.8789	0.9117	0.8953
		70:30	0.8807	0.9384	0.8560	0.8972	0.8957	0.8796	0.9026	0.8911
500	Original	90:10	0.9106	0.1074	0.9613	0.5536	0.9410	0.5997	0.9789	0.7893
		80:20	0.8995	0.6059	0.9730	0.7894	0.9063	0.7357	0.9490	0.8423
		70:30	0.9009	0.8372	0.9281	0.8827	0.9012	0.8297	0.9318	0.8807
	RUS	90:10	0.8484	0.9383	0.8384	0.8883	0.8795	0.9121	0.8758	0.8940
		80:20	0.8552	0.9406	0.8339	0.8872	0.8786	0.9034	0.8724	0.8879
		70:30	0.8791	0.9471	0.8499	0.8985	0.8926	0.9047	0.8874	0.8960
	ROS	90:10	0.8532	0.9445	0.8430	0.8938	0.8890	0.9073	0.8869	0.8971
		80:20	0.8574	0.9425	0.8361	0.8893	0.8847	0.9072	0.8790	0.8931
		70:30	0.8762	0.9474	0.8457	0.8966	0.8914	0.9069	0.8847	0.8958
	SMOTE	90:10	0.8524	0.9449	0.8422	0.8935	0.8977	0.8913	0.8984	0.8949
		80:20	0.8593	0.9456	0.8378	0.8917	0.8887	0.8934	0.8876	0.8905
		70:30	0.8824	0.9386	0.8584	0.8985	0.8976	0.8894	0.9010	0.8952
	ANS	90:10	0.8557	0.9420	0.8461	0.8940	0.9014	0.8852	0.9032	0.8942
		80:20	0.8593	0.9453	0.8378	0.8915	0.8899	0.8882	0.8903	0.8893
		70:30	0.8822	0.9381	0.8582	0.8982	0.8982	0.8863	0.9034	0.8948
	DBSMOTE	90:10	0.8568	0.9360	0.8480	0.8920	0.9116	0.8563	0.9178	0.8870
		80:20	0.8626	0.9375	0.8439	0.8907	0.8968	0.8579	0.9065	0.8822
		70:30	0.8824	0.9359	0.8595	0.8977	0.9009	0.8694	0.9144	0.8919

Table II. (contd.)

n	Handling Technique	IR	MLE				Score			
			Acc	Sens	Spec	BA	Acc	Sens	Spec	BA
1,000	Original	90:10	0.9222	0.2232	0.9438	0.6115	0.9402	0.5842	0.9798	0.7820
		80:20	0.8970	0.5684	0.9791	0.7738	0.9037	0.7095	0.9522	0.8309
		70:30	0.8978	0.8370	0.9238	0.8804	0.9063	0.7357	0.9490	0.8423
	RUS	90:10	0.8479	0.9557	0.8359	0.8958	0.8824	0.9134	0.8790	0.8962
		80:20	0.8569	0.9301	0.8386	0.8843	0.8761	0.8844	0.8741	0.8793
		70:30	0.8715	0.9469	0.8391	0.8930	0.8882	0.9110	0.8784	0.8947
	ROS	90:10	0.8518	0.9539	0.8405	0.8972	0.8838	0.9134	0.8805	0.8970
		80:20	0.8569	0.9422	0.8356	0.8889	0.8771	0.8966	0.8722	0.8844
		70:30	0.8725	0.9465	0.8407	0.8936	0.8913	0.9107	0.8830	0.8969
	SMOTE	90:10	0.8495	0.9578	0.8375	0.8976	0.8920	0.9044	0.8906	0.8975
		80:20	0.8545	0.9301	0.8356	0.8828	0.8837	0.8883	0.8825	0.8854
		70:30	0.8782	0.9362	0.8533	0.8948	0.8945	0.8977	0.8931	0.8954
	ANS	90:10	0.8506	0.9557	0.8390	0.8973	0.8947	0.8993	0.8942	0.8968
		80:20	0.8537	0.9301	0.8346	0.8823	0.8829	0.8880	0.8816	0.8848
		70:30	0.8785	0.9364	0.8537	0.8950	0.8950	0.8939	0.8955	0.8947
	DBSMOTE	90:10	0.8508	0.9484	0.8399	0.8942	0.9081	0.8675	0.9126	0.8901
		80:20	0.8602	0.9380	0.8408	0.8894	0.8879	0.8588	0.8952	0.8770
		70:30	0.8789	0.9339	0.8553	0.8946	0.8977	0.8806	0.9051	0.8928